

Monsoon Semester, 2018, Faculty of Architecture, CEPT University

Code	1117	Course	History- III		
Typology	Courses	Credits	2	GPA/NonGPA	NonGPA

Long Description :

This lecture based advanced history of architecture course will present an overview of the key concepts and changes that occurred in architecture from the era of industrialization to the present. The impact of industrialization and colonization, the advent of Modernism around the globe will constitute the content of this course.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Emphasises architectural developments in the context of industrialisation and modernisation
- Focus on understanding modernity as experience and as a intellectual, social, economic and political paradigm
- Focus on the nature of Indian modernity vis-a-vis other modernities

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Theme 1: Modernity and the Enlightenment Project
Week 2	Industrial Revolution: Technological & Cultural Changes in Europe
Week 3	Transformation of cities and emergence of new building types
Week 4	Students presentations: Global perspective on Theme 1
Week 5	Theme 2: Colonialism
Week 6	Colonialization and modernities - a new world order
Week 7	Architectural, technological, and aesthetic legacies
Week 8	Student presentations: Global perspective on theme 2
Week 9	Theme 3: Towards Modernism; A history of architecture as a profession
Week 10	Modernism as a intellectual, stylistic and cultural movement
Week 11	Indian Modernity: Colonial and nationalist trajectories
Week 12	Student presentations: Global perspective on theme 3
Week 13	Theme 4: Is there an Indian architecture?
Week 14	Indian Architecture Post- Independence: Legacies of PWD, Institutions and Nation Building Work of Early Modern Indian Architects
Week 15	The Post War World - after 1950
Week 16	Student presentations: Global perspective on theme 4

Monsoon Semester, 2018, Faculty of Architecture, CEPT University

Code	1147	Course	Surface (Flying!) Structures		
Typology	Courses	Credits	2	GPA/NonGPA	NonGPA

Long Description :

tbd

Learning Outcomes :

After completing the Courses, the student will be able to :

- Student would be able to : intelligently discuss large span structural systems develop conceptual ideas for such space problems detail to a basic extent the structural system understand how to respond to structural issues in an aesthetic manner. Develop representational models / drawings / sketches of such systems.

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Introduction and Types of surface structures.
Week 2	Structural behavior, membrane, shell, flexural systems case studies, introduction to assignment.
Week 3	Presentation of assignments.
Week 4	Form finding methods, introduction to membrane structures.
Week 5	Preparation of models
Week 6	Introduction to grid structures and grid systems, making models using low E materials
Week 7	Importance of detailing and drafting out of ideas of structural system. Understanding the need and use of analytical methods in addition to intuitive methods for detailing.
Week 8	Analytical methods for shells, membranes and grid structural systems using programs like rhino, grasshopper karamba SAP2000 etc. Final assignment introduction
Week 9	Discussions for final assignment
Week 10	Discussions for final assignment
Week 11	Submissions and juries of final assignment
Week 12	Submissions and juries of final assignment
Week 13	No Lecture
Week 14	No Lecture
Week 15	No Lecture
Week 16	No Lecture

Monsoon Semester, 2018, Faculty of Planning, CEPT University

Code	4653	Course	Transforming Cities: Making City Planning and Urban Design Work in India		
Typology	Lecture	Credits	2	GPA/NonGPA	NonGPA

Long Description :

This course will present key urban planning and design projects lead by Prof. Bimal Patel over the last 25 years. Students will learn how these projects came into being, why they were designed as they were, how they were implemented and how they relate to larger issues in urban planning, urban design and urban development. The aim is to focus on lessons learned so as to equip you to be effective practicing planners and urban designers. Introduction and objective of the course Urbanization presents an epochal opportunity to intertwine economic growth with social and human development. It also poses many formidable challenges and yet, since it takes place only once, no country can afford to squander the opportunity that it presents. Effective urban planning and urban design, focused on transforming our cities to be more livable, efficient and sustainable, are key to seizing this opportunity and tackling the challenges of contemporary urbanization. Over the last twenty-five years Prof. Patel and his colleagues at EPC and HCP have worked on many pioneering urban planning and urban design projects aimed at transforming and improving Indian cities. Their work has involved imagining what Indian cities can be, planning them, writing legislations, analyzing and framing policies and regulations, advocating reforms, designing public spaces, designing urban infrastructure and managing complex urban development projects. This course will present their work as a series of case studies. The lectures will present the projects, the strategies that were used and the lessons that were learned. Each lecture will be followed by a discussion session. Learning objectives Students will learn how to gather data about urban projects and how to analyze them. They will develop an understanding of the genesis of various urban projects, and the different stages they then go through. This course ultimately offers students the opportunity to reflect upon a planner's or designer's scope and role in the creation and implementation of a project.

Learning Outcomes :

After completing the Courses, the student will be able to :

- will be circulated later

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	1. Challenges and Opportunities of Urbanization What is urbanization? Why do towns and cities fall apart under the onslaught of urbanization? To what extent is India unique? How do elites react to urbanization? Is urbanization reversible? What threat and opportunities does urbanization pose? What is important in tackling the challenges and seizing the opportunities that urbanization poses? What roles can urban planning and urban design play? Why is this course structured around projects and what are its objectives?
Week 2	2. The Mumbai Eastern Waterfront Development Mumbai's port activities have shifted from Mumbai Port's old location, along the Island City's eastern waterfront, to Nahva Sheva, which is across the bay. Much of the port's 7 sq. km. of land is lying unutilized or underutilized. This lecture will describe the design for redevelopment of the Mumbai Port Trust's land and discuss the many objectives that the redevelopment will serve, from rejuvenating Mumbai's economy, creating vast new amenities for Mumbai's citizens, to unlocking the value of the port's land. It will discuss the many political-economy issues that planners have to deal with in such large projects. It will also discuss the urban morphology that the redevelopment plans to use, which based on early twentieth century models of urban development in India and radically different from the current norm.
Week 3	3. The C G Road Development This tiny street redevelopment, undertaken in the mid-nineties, was one of the first designed public space improvement project in recent times, in the country. It introduced a number of novel ideas to Indian urban development in the mid-nineties: the idea that streets should be designed, the idea that urban development projects could be self-financed; the idea that urban development projects could be public-private partnerships, and the idea that cities should create well-organized and paid on-street parking. The lecture, besides briefly discussing the design of the street, will focus on the many implementation challenges that were faced, how they were overcome and what lessons can be learned from them. It will also cover some subsequent work on street design and development.
Week 4	4. The Ahmedabad and Surat Walled City Plans Undertaken in the late nineties, these plans were aimed at revitalizing the walled cities in Ahmedabad. The planners worked at facilitators, helping staff in various departments of the municipal corporations come up with a comprehensive roster of carefully thought-out, sufficiently detailed and well-interlinked projects. The projects themselves were of various types. The underlying objective of both plans was to demonstrate the power of systematic, comprehensive and consultative planning during a period when haphazard, disjointed and personalized investment planning and policy making was more the norm. The lecture will use these projects to discuss the functions of urban planning. What can urban planning hope to achieve? What can the role of a planner be in local governments? What can the role of consultants be in urban planning? What are planners up against when they try to plan within the context of local governments?

Week 5	5. The Development Plan for Ahmedabad, 2002 Ahmedabad's first citywide development plan was made in 1965. It made many bold proposals but they proved difficult to implement. A decade later, when the plan was due for revision, it was clear that its proposals had failed. However, regardless of the failure of earlier proposals, Ahmedabad's plan was revised using the same philosophy that had informed the 1965 plan. Unsurprisingly, the revised 1976 plan did not fare better than the earlier one. Yet, once again, in 1986, the 1976 plan was revised using the same old approach. For another decade Ahmedabad's plan remained unimplemented and the city continued to grow without reference to it. Fortunately, by the late nineties, when the plan was again due for revision, the political climate had changed. The new plan, sanctioned in 2002, was informed by a new liberal paradigm. Abandoning the old approach to planning proved to be very beneficial. It was now possible to implement the plans regulations and proposals - planning began to work. This lecture will discuss the making of this new plan. What were the key new features of this plan? What were the structural and contingent reason's that made it possible to adopt a new approach to planning in turn of the century Ahmedabad? What is meant in a liberal approach to planning?
Week 6	6. Restructuring Bhuj Walled City, 2004 On 26th of January 2001 Bhuj was devastated by a severe earthquake. A large proportion of buildings in its dense, one sq. km. walled city either fell down, or were damaged. By the time that the debris was cleared, almost 60 percent of the buildings were no longer there. During the earthquake many people had died because there was no where to run and, because the narrow street were blocked by debris of fallen buildings making it impossible for rescue vehicles to access most areas within the walled city. As a consequence, when plans were being drawn up to rebuild infrastructure in the walled city, many residents expressed a desire to widen some of the streets to improve access into the dense fabric of the walled city. Regardless of the depth of this desire among most residents the government knew that widening streets by acquiring land along old street alignments would prove politically impossible - given the circumstances, no one would be willing to bear the cost of creating public benefits by making private sacrifices. It was therefore decided to build three brand new loop streets through areas that had been laid open by the earthquake's devastation. It was also decided to use the T P Scheme mechanism to appropriate land for the streets. This made it imperative to restructure the plot layout of almost all the approximately 12000 plots in the walled city. This lecture will use the restructuring of the walled city to discuss various aspects of TP Schemes, more generically know as Land Pooling/Land Reconstitution. Why does land pooling work? Why is it not widely adopted? What can be done to expand its use?
Week 7	7. Ahmedabad and Gandhinagar's innovative Town Planning Schemes Ahmedabad's first Town Planning Scheme was prepared in 1925. By now over one hundred schemes have been prepared to manage planned growth of Ahmedabad and Gandhinagar. This year also marks a century since the invention of this extremely important planning, plan implementation and plan financing tool. Most urban planner tend to see TP Schemes as a tool for managing peripheral urban growth. They do not conceive of it as a versatile tool that can be used for achieving many other planning objectives. This lecture will focus on town planning schemes - in Thaltej, Koba, (illegal,) South Ahmedabad, Sanskardham, and along the Ring Road where T P Schemes have been used for novel purposes. It will also discuss some speculative projects where T P Schemes themselves have been innovated upon.
Week 8	8. Public Land Study This study was undertaken with World Bank support and published as a World Bank Working Paper. It asked two simple questions: How much public land does Ahmedabad have? What is the market value of the unutilized portion of this land? Though the initial questions were simple they raised a whole host of ancillary questions and many methodological issues. What do we mean when we say 'public land'? Given that land records are practically inaccessible in India, is it at all possible to build an inventory of public land? How can one calculate how much of a plot's development potential has been utilized? How does one assess the market value of plots distributed across the whole city in a cost effective manner? Very innovative methods were developed to answer all these questions. This lecture will focus on the Public Land Study to discuss more general issues pertaining to public land and planning research. Why do Indian cities have so much underutilized public land? How can it be better utilized? How can the hurdles to useful planning research be overcome? With reference to this last question the lecture will also discuss the project to inventory Mumbai's environmental assets.

Week 9	9. The Justice J U Mehta Committee Report In 2000, Justice B C Patel of the Gujarat High Court launched a crusade against rampant illegal construction in Ahmedabad. Through a series of suo moto injunctions he forced the AMC and the city at large to take cognizance of the fact that a vast proportion of new construction in the city was illegal and therefore liable to being demolished. He also forced some high profile demolitions and forced residents in many high-rise apartment buildings to install fire safety equipment that, according to the prevailing building regulations was required, but had not been provided. He then established the Justice J U Mehta Committee, to investigate illegal constructions in Ahmedabad and to pin specific responsibility individual municipal officers for allowing illegal buildings to come up. Based on the request of one of the committee members, Dr. Bimal Patel, the Committee's scope of work was expanded to include a study of the building regulations system in Ahmedabad to identify its shortcomings and to suggest remedies to improve it. This lecture will also discuss the work of other initiatives that Dr. Patel has been involved in to improve compliance to building regulations in Ahmedabad and elsewhere. The key questions that this lecture will deal with are: What are the reasons underpinning widespread non-compliance to building regulations? What are the qualities of a 'good' building regulations system? What will it take to enforce buildings regulations in India?
Week 10	10. Kankaria Lakefront Development, Kankaria is a lake in southeast Ahmedabad developed by Sultan Qutbuddin in the 15th century. When, in 1924, AMC developed the Maninagar residential suburb in the lake's vicinity, the lake was ringed by a circular street and surrounded by a number of public facilities. They included a zoo, a garden, a children's amusement park, a public swimming pool and some other municipal facilities. Until the early 1990s, Kankaria and the public facilities surrounding it served Maninagar and Ahmedabad well as a very pleasant and enjoyable recreational asset within the city. However, by the turn of the century, with increasing traffic, more and more unregulated parking and vending and poor upkeep, Kanakaria's value as a recreational spot had considerably diminished. It is at this time that, at the urging of the then Chief Minister Modi, the AMC undertook redevelopment of Kankaria. Redeveloped Kankaria became a highly popular spot in the city and restored its status as an Ahmedabad icon. It also changed its character as a public space. Except in the mornings, when it continued to be an open neighborhood public space, it became a gated, ticketed, pedestrianized, amusement facility. At the time when the redevelopment was underway, this change was strongly opposed by many left leaning civil society organizations who argued that the design violated fundamental norms pertaining to public spaces. This lecture, besides describing the process and design of the redevelopment will focus on a number of questions that it raised. What are public spaces? How should the character of a public space be defined? What were the clashing political-economic views that the redevelopment brought to the fore? Is the role of a professional planner when such contesting views have to be dealt with?
Week 11	11. Ahmedabad Housing Studies Housing strategies in traditional urban plans are based on the 'predict and provide' approach. Planners attempt to 'predict' future demand and expect the government to 'provide' housing to the urban population. They 'reserve' land for housing for those who cannot afford it and expect government to acquire it and expect government housing agencies to build affordable housing on the acquired land. This approach has never been effective. Though many planners agree that it should be replaced by a more realistic market based approach, most are unclear about what this new approach should be. To answer this question, the World Bank supported a set of studies to analyze the supply and demand for housing in Ahmedabad. The team that worked on this research project housed at Environmental Planning Collaborative included several national and international experts. This lecture will discuss this study, as well as some others; to flesh out what a market based approach to housing would look like.
Week 12	12. The Ahmedabad CBD Plan With, rising populations and increasing prosperity demand for floor space in almost all Indian towns and cities is bound to increase exponentially. This, in turn, is going to result in vast swathes of built up areas in towns and cities being redeveloped - building by building, as small buildings are torn down to make way for larger buildings. Though such redevelopment is already underway in most towns and cities none, bar Ahmedabad, is focused on systematically tackling the challenges that piecemeal redevelopment poses and the incredible opportunities that it presents. This lecture will focus on Ahmedabad's plan for the redevelopment of its Central Business District. It will explain the plan's logic and approach of to show why it has the potential of transforming the discourse around urban redevelopment. Today, it is rightly believed that if the demand for more floor space (expressed through rising land prices) is met by increasing FSI limits, real estate developers and landowners will gain, and ordinary people will loose out on account of stressed infrastructure and increasing traffic congestion. Can urban planning counter this real threat and turn urban redevelopment into a win-win proposition? What qualities must an implementable urban redevelopment plan have?

Week 13	13. The Sabarmati Riverfront Development Project - Project Design and Development Management Most rivers passing thorough Indian cities are severely polluted, encroached upon by slums and inaccessible. During the early nineties the Sabarmati in Ahmedabad was in such a state. Then, in the mid-nineties, the Ahmedabad Municipal Corporation formulated an ambitious plan to clean up the river and transform 11km of the riverfront into a multipurpose public space. Key components of the project were realized over the next 15 years. Others are still underway. The project is widely recognized to be a pioneering urban development project. It is also widely admired for its scope, its scale and for its successful implementation. This first lecture on the project will discuss the design of the Sabarmati Riverfront Development Project and the unusual manner in which the project was managed during it crucial early phase. Why was the project designed the way it was? What aspects of the design were key to its successful implementation? What were the management challenges that the project faced? How were they overcome?
Week 14	14. The Sabarmati Riverfront Development Project - Relocation and Rehabilitation and Engaging the Public Most rivers passing thorough Indian cities are severely polluted, encroached upon by slums and inaccessible. During the early nineties the Sabarmati in Ahmedabad was in such a state. Then, in the mid-nineties, the Ahmedabad Municipal Corporation formulated an ambitious plan to clean up the river and transform 11km of the riverfront into a multipurpose public space. Key components of the project were realized over the next 15 years. Others are still underway. The project is widely recognized to be a pioneering urban development project. It is also widely admired for its scope, its scale and for its successful implementation. This second lecture on the project will discuss the relocation of 11000 families from slums along the riverfront to new housing within the city. It will also discuss the communications strategies that were employed to explain the project to the public.
Week 15	15. Making City Planning and Urban Design Work in India This concluding lecture will discuss how city planning and urban design can be more effective in tackling the peculiar challenges that India poses – a raucous and disruptive democracy, a debilitating bureaucracy and a severe lack of resources. What approaches and strategies can planners and urban designers use to be more effective in transforming and improving Indian cities?
Week 16	Discussion

Spring Semester, 2019, Faculty of Architecture, CEPT University

Code	AR2016	Course	Tectonic Operations		
Typology	Studio	Credits	14	GPA/NonGPA	GPA

Long Description :

Studio explores ways through which outdated buildings can become sites for creative thinking about construction processes. It shall begin by introducing new programmatic requirements in an old precinct, necessitating actions of repair, replacement or introduction of alternative parts. This'll anchor the play of new technologies and materials into clearly identified spatial and functional conditions rather than being based merely on personal whim or random choice. This may be understood through a biological analogy of Processes, Uses, Form & Location and Materials. Through this, new formal, structural & material conditions will be explored for finding unique architectural expression.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Visualize existing buildings from the point of view of structural possibilities & scope for intervention.
- Develop skills for identifying building language and developing its grammar through detailing out architectural elements
- Undertake a small renovation project, with knowledge of different agencies that get involved, the stages & sequences of construction.

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	SITE STUDY Detailed mapping of existing structures, materials & uses Exploration of new possible uses. Identifying case studies, projects which integrate new uses and forms in desirable ways.
Week 2	SITE STUDY Detailed mapping of existing structures, materials & uses Exploration of new possible uses. Identifying case studies, projects which integrate new uses and forms in desirable ways.
Week 3	preparing the revised schema using drawings and models to integrate new functional requirements, their geometries and elements with existing structures. clarification of structural & construction techniques and approach
Week 4	preparing the revised schema using drawings and models to integrate new functional requirements, their geometries and elements with existing structures. clarification of structural & construction techniques and approach
Week 5	WALL SECTION STUDIES detailed cutaway axonometric drawings of the new proposal at scale 1:20
Week 6	WALL SECTION STUDIES detailed cutaway axonometric drawings of the new proposal at scale 1:20
Week 7	DETAIL STUDY OF STRUCTURAL SYSTEMS foundation / plinth floor (deck, mezzanine) vertical linear/planer elements (walls, partitions, columns, etc.) roofs (pitched, tensile, folded plates, vaults, shells, etc.) AT LEAST 2 OF THE ABOVE CIRCULATION SYSTEM stairs, ramps, elevators, etc. AT LEAST 1 OF THE ABOVE
Week 8	DETAIL STUDY OF STRUCTURAL SYSTEMS foundation / plinth floor (deck, mezzanine) vertical linear/planer elements (walls, partitions, columns, etc.) roofs (pitched, tensile, folded plates, vaults, shells, etc.) AT LEAST 2 OF THE ABOVE CIRCULATION SYSTEM stairs, ramps, elevators, etc. AT LEAST 1 OF THE ABOVE
Week 9	DETAIL STUDY OF STRUCTURAL SYSTEMS foundation / plinth floor (deck, mezzanine) vertical linear/planer elements (walls, partitions, columns, etc.) roofs (pitched, tensile, folded plates, vaults, shells, etc.) AT LEAST 2 OF THE ABOVE CIRCULATION SYSTEM stairs, ramps, elevators, etc. AT LEAST 1 OF THE ABOVE
Week 10	DETAIL STUDY OF STRUCTURAL SYSTEMS foundation / plinth floor (deck, mezzanine) vertical linear/planer elements (walls, partitions, columns, etc.) roofs (pitched, tensile, folded plates, vaults, shells, etc.) AT LEAST 2 OF THE ABOVE CIRCULATION SYSTEM stairs, ramps, elevators, etc. AT LEAST 1 OF THE ABOVE
Week 11	DETAIL STUDY OF STRUCTURAL SYSTEMS foundation / plinth floor (deck, mezzanine) vertical linear/planer elements (walls, partitions, columns, etc.) roofs (pitched, tensile, folded plates, vaults, shells, etc.) AT LEAST 2 OF THE ABOVE CIRCULATION SYSTEM stairs, ramps, elevators, etc. AT LEAST 1 OF THE ABOVE

Week 12	DETAIL STUDY OF SERVICES Toilets Kitchen / pantry / yard Pools / other operational water bodies AT LEAST 2 OF THE ABOVE CLIMATE RESPONSE Doors / windows Screens / weather shading devices AT LEAST 1 OF THE ABOVE
Week 13	DETAIL STUDY OF SERVICES Toilets Kitchen / pantry / yard Pools / other operational water bodies AT LEAST 2 OF THE ABOVE CLIMATE RESPONSE Doors / windows Screens / weather shading devices AT LEAST 1 OF THE ABOVE
Week 14	DETAIL STUDY OF SERVICES Toilets Kitchen / pantry / yard Pools / other operational water bodies AT LEAST 2 OF THE ABOVE CLIMATE RESPONSE Doors / windows Screens / weather shading devices AT LEAST 1 OF THE ABOVE
Week 15	STUDY OF TACTILE CONDITIONS Studying articulation or continuity of surfaces, edges & corners STUDY OF Lighting Design Flooring Cladding / plasters other surfaces applications & Colour study AT LEAST 2 OF THE ABOVE
Week 16	STUDY OF TACTILE CONDITIONS Studying articulation or continuity of surfaces, edges & corners STUDY OF Lighting Design Flooring Cladding / plasters other surfaces applications & Colour study AT LEAST 2 OF THE ABOVE Juries Portfolio Grading and Exhibition

Spring Semester, 2019, Faculty of Architecture, CEPT University

Code	AR2603	Course	Contemporary Directions in Architecture		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

Input lectures on specific current issues like sustainability / environmental friendly architecture, digital architecture, public architecture and conservation will include theoretical premises and examples of actual built works. The input sessions will be followed by student led discussions in various formats. The students are expected to read, analyse and then articulate their learning through various forms of communication. This course designed for senior students will bring to their learning current dilemmas of the profession, both in theory and practice.

Learning Outcomes :

After completing the Courses, the student will be able to :

- The students will be able to articulate the plurality in the expressions of architecture in contemporary times.
- The students will be able to analyse specific texts to articulate the arguments presented.

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	The question of history and the discontent with modernism – emergence of a plurality in architecture.
Week 2	Readings (list at the end) in architectural theory that determined the deviation from the dominant Modern Architecture will be discussed and the context understood.
Week 3	Readings (list at the end) in architectural theory that determined the deviation from the dominant Modern Architecture will be discussed and the context understood.
Week 4	The changing notion of public architecture - Institutions, public spaces
Week 5	The changing notion of public architecture - Housing
Week 6	Theoretical premise of the environmental concerns: the history and the various theories that are grounding the present context.
Week 7	Environmental concerns - various approaches
Week 8	Conservation – various positions, debates, theories.
Week 9	Approaches to conservation: various examples.
Week 10	Digital architecture - possibilities and limitations
Week 11	Readings (list at the end) in architectural theory that determined the deviation from the dominant Modern Architecture will be discussed and the context understood.
Week 12	Final quiz
Week 13	No classes
Week 14	No classes
Week 15	No classes
Week 16	No classes

Spring Semester, 2019, Faculty of Architecture, CEPT University

Code	AR2705	Course	Material Adventures		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

Today there is a vast variety of materials available and moreover claims made by patented materials makes really difficult to make the right choice of materials. Through this course, we should take an approach to understand the basics of materials and their properties. While discussing the material classification, would go for in depth analysis of properties as well as would discuss the bonding/joining issues and design with such materials. References would be made to appropriate applications. Along with properties and application technics the burning issues will be resource value, embodied energy and operational energy as well. A life cycle analysis will help us draw complete picture about the materials. The whole journey will be covered through theory and experiences. Almost 40-50% time would be class room discussions. Rest will be industrial visits and hands on experience with the materials. The group/individual assignment would be working with materials and life cycle analysis. There may be long distance site visit on Saturday.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Appreciating classification of material and properties represented by the group
- Understanding the effects and characteristics impacted by the manufacturing process.
- Hands on experience will teach importance of skill and their efforts required in fine working methodology
- General physical properties and testing there off.

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Material Classification and properties representing the classification done.
Week 2	Ceramic, mud, lime and cement
Week 3	Industrial visit
Week 4	Laboratory testing
Week 5	Renewable Materials and renewable products
Week 6	Workshop based exercise
Week 7	Metal
Week 8	Polymer and composite
Week 9	Site Visit
Week 10	Assignment Presentation
Week 11	Embodied Energy and Life Cycle analysis
Week 12	Concluding session

Spring Semester, 2019, Faculty of Architecture, CEPT University

Code	AR2702	Course	Form, Material and Structure		
Typology	Courses	Credits	2	GPA/NonGPA	NonGPA

Long Description :

Form, material and structure - how they are selected, manipulated or associated in design produces the outcome, therefore the understanding of form, material and structure is very important for designers. Material properties and structural actions are vital for development of form. The course begins with developing the understanding of structural actions, properties of structural materials and form which will lead to deeper understanding of interrelationships of these three ingredients. The course will be conducted as lecture presentations interspersed with relevant model making assignments

Learning Outcomes :

After completing the Courses, the student will be able to :

- Students will gain the basic understanding of structural actions
- Students will develop an ability to find the appropriate structural forms based on structural actions and material properties
- Students will develop an ability to decipher the structural system in the given building

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Basic understanding of form, material & system
Week 2	Basic understanding of form, material & system
Week 3	Structural materials
Week 4	Structural materials
Week 5	Form finding
Week 6	Form finding
Week 7	Structural systems
Week 8	Structural Systems
Week 9	Application of understanding of form, material & system – Large span structures
Week 10	Presentation
Week 11	Presentation
Week 12	Concluding Session
Week 13	-
Week 14	-
Week 15	-
Week 16	-